



De Leon Melendez, Jesus Manuel
Manager: Watson, James E - Sr Mgr-Radio Frequency

2022 Year End

Sr Engr CsIt-Radio Frequency

Best Network

1. Consistent monthly reporting/feedback to the regions on priorities.
 - a. Include SP/RF directors
 - b. Review of alignment of POR/BPY dates to KITT goals
 - c. Constant reinforcement of status and BPY/POR definitions, and the need to keep all dates constantly maintained
2. Monitor changes to network traffic and report out on it (requires executive, territory, and market-level reports as well as detail level results for the submarkets to use and take action on; target end of 1Q to begin sharing reports, 2Q for the more difficult reports)
 - a. C-Band
 - Offload
 - Total Data Volume by PEA or LTE or Submarket
 - % of total PEA data traffic - 4G, NW, C-Band, mmWave
 - Analysis of per site information - C-Band as a % of total site traffic, trigger relief on congested sites, data volume increases tied to congestion relief (as best as we can see)
 - UEs per sector or site, max UEs, unique UEs
 - Throughput analysis
 - b. mmWave
 - Offload
 - Analyze need for 4G/5G combo sites
 - Support Waypoint
 - c. 5G versus 4G Analysis
 - Traffic stats designed to highlight the trends and shifts of 4G traffic to 5G NW and 5G mmWave
 - Trigger relief as a result of 5G adoption/usage
 - Data volume trends as a result of relieving congestion

- d. QCI8 and QCI9 Traffic Studies
 - Monitor Changes, especially when price plans change
 - e. Busy Hour
 - f. FWA (includes OFS, LTE Home, 5G home)
 - User and sector information on 4G FWA
 - Usage / node or site, usage / customer, time of day usage stats, % of sector/site DV consumed by fixed products, number of fixed users per sectors/nodes, throughput trends
 - User and sector information on 5G FWA
 - Same as above but also broken down by 5g technology (NW, C-Band, mow)
 - g. CDMA sunset
 - CDMA traffic by cell/ RSA/ submarket broken down in actual volume, % of total traffic, etc
 - MDN analysis of cdma only users as well as dual mode devices with high cdma volume
 - 3. Trigger 4G and 5G (part of monthly report 1 listed above)
 - a. What sector(s) need a solution?
 - b. What solutions need to be released?
 - c. Which solutions are released and need to be prioritized?
 - d. "Top 10' List per Submarket
 - 4. Continue to drive and support the territory teams through the project management of assigned key programs and core teams.
 - a. Special projects include: C-Band, mmWave, SNAP, ROOT, RBB, SGFI, STL, and more
 - b. C-band - Work with field teams to prioritize sites for maximum POPs count to drive toward 200M POPs covered by end 2022, 23K LTD, and 4368 C-Band builds outside the initial 46 PEAs
 - c. mmWave expansion - 45k cumulative sites, 12k new sites
 - d. Readout on your projects regularly to keep the team up to speed; request to join RF or Engineering director calls as needed to share updates
 - 5. Network Planning
 - a. 3 year Plan - support Vision and submarket pipelines
 - b. Assist with the development of the 2023 plan
 - 6. Ensure new sites are being deployed in the right areas and are solving the issue they were intended to solve (SGFI, Capacity, etc)
 - 7. C-band - Continue to drive and support the C-band program. Work with field teams to prioritize sites for maximum POPs count to drive toward 200M POPs covered by end 2022.
-

I've directly supported the regions by having weekly meetings with them where we discuss the latest national announcements and guidelines to ensure a one voice approach. I've worked with them on solution releases, rationale corrections, project dissassociations, several data audits and solution priority. Currently I hold a monthly capacity meeting for UPNY to evaluate solution needs and guide them through sectors that need solutions. There is an opportunity to

integrate this format to NYM as their triggers seems to be going up as traffic seems to be returning to levels previous pandemic. For now new build solution release have been scarce as the solutions list are full when taken into consideration the pipeline to funding ratio. This last part changed at the end of the year, with the business decision to give back control of the pipeline to the Regions. I've pivot my focus into emphasizing importance of solution priority and education on how its calculated. Ive develop knime workflows for both KSS and KSP values to be able to easily check these in NPP as we have been having issues during the last year. I worked on the team was able to stabilize KSP and find the root cause of the issues. Ive also spend a good amount of time pushing for the correction of the KSS values which still yet to be fixed.

De Leon Melendez, Jesus Manuel 1/23/2023

Best Systems

1. Support Core team activities
 - a. Timely submit feature requests for upcoming PIs
 - b. Submit and escalate bug reports and tickets as needed
 - c. OPT and APT development - continue to engage with HQ and the developers on these tools and the algorithms that drive them (support ASEU forecasting model, KSS, KSP, etc)
 - d. Data Integrity – find errors, create audits, work with the submarkets to clean up our tools and golden sources , submit feature requests to fix issues
 2. N7L - supporting testing and reporting bugs
 - a. Support incorporation of seasonality
 3. Support SAFE process and prioritize new features
 4. Ensure alignment and clear communication with the finance team (especially when we make changes to Primary Rationale, which change the budget codes)
 5. Become familiar with 1ERP (high level is sufficient)
-

As part of the APT team I help developed the new Kracken revenue target (w Michael Neidt) for the new N7L data configuration. Ive also participated in the testing for data ingestion and quality for the 5G tables that will make it eventually to APT. Ive also lead the meeting on a couple of occasions while covering vacations for our host.

Created NPP (now CANVAS) feature request for capacity solution evaluation pre/post report.

I created the new tables for Kracken Revenue targets for the N7L configuration that are used to determine the KRW applied to KSP values.

At the end of the year I worked in the prioritization revue for enodeBsectorGrps. I created a dashboard to visualize and compare the proposed new formulas. Work is still ongoing, but we should have a feature request early in the year.

I've also develop the Program Label dashboard to automate Sector capacity labels based on the criteria developed by our team, simplifng the task for our team so we can be more productive with our time. End result is a downloadable file our engineers can use to vlookup the solution labels that apply.

On of the most succesful visualizations I created last year was the APT trigger trend. It shows the user trigger trend by different dimensions linearly and year over year. It has been specially usefull with the refocus on trigger rates and has been driving more users to our dashboard page.

As DSS to clean 5G NR carrier was being discussed nationwide I've help develop a visualization for full 850 NR capacity to forecast triggers based on the limited new capacity and to show how we were overstating the sub1 capacity at the time. This helped drive the discussion of the involved teams so an agreement can be made and expectations set with our leadership.

As part of the data integrity team I've helped advise the 30 day running audit which will help us better monitor possible data issues. I've also developed a dashboard to compare budget codes in NPP and SPM to drive the correction of mismatches between systems before 1 ERP is integrated

De Leon Melendez, Jesus Manuel 1/23/2023

Best Cost

1. Pipeline/Pool refresh (part of monthly report)
 2. Audits – are the RFDS designs matching the RF design standards?
 3. Work with Finance
 4. Support the TO Savings Initiatives
-

During the year I worked on HRR with Shellie Ruiz team to ensure regions have a healthy pipeline doing project creations and releases on a national lever. So far we have released a total of 1861 projects to the region. By the end of the year we have released 5600 new Relos which should help drive the cost of the business down. I've also audited for Rationale mismatches and projects that are not in the teams purview.

I also constantly work on audits with my regions to clean up databases and ensure our data is the best it can be. Although we are not officially doing a pipeline refresh I do pushback on solutions that are lower priority and try to hold a region to a reasonable ammount of solutions.

I took the opportunity this year I asked Sunil John and Kyle Smith to cooperate with their team and became part of the team working on Solution effectiveness report as a traffic liaison, giving advise on how to account for capacity and handle small cell clusters among other topics. We have been meeting weekly during the past year to develop this tool which should be publish to market on the 1st quarter of 2023.

De Leon Melendez, Jesus Manuel 1/30/2023

Best Culture

1. Mentoring and/or teaching others. Once a quarter to provide a team training (can be on team call or separate call) on a topic to help educate or inform others. Can be a traffic related topic, about a tool (how to better use Tableau for example), or another work related topic.
2. Drive a curious learning and collaborative culture
3. Improve work processes. Find creative ways to improve current work processes.
4. Volunteer activity. Find ways to get the team engaged in a volunteering activitiy.

- 5. Improve Communications
 - a. Active participant in regular weekly or bi-weekly meeting with region.
 - b. Work with team members to provide a once a month review of Traffic monthly report
- 6. Take mandatory training (driver training, etc.).

Within the Traffic team I try to apply the Credo by being a positive agent and helping my teammates anyway I can. I've Worked with Abdul developing the 5G UWB combo dashboard and on the logic for the small cell opportunities based on the RF Standards team guidance. Updated the Solutions insider Dot Metrics dashboard and added Automatic classification based on SPM project coordinates. I've also supported Irene Fernandez in the development of dashboards such as the most broken sector and the executive dashboard to provide an easier insight to our team and regions based on our KPIs.

I think I am a good asset to the team by bringing good energy to work, having a cooperative mind and take time to share knowledge with anyone to ask for it. This year we have had multiple new additions to the team, which have allowed me to mentor one on one about different aspects of our tools. Ive become one of the go to resources for dashboard creation and data visualizatio.

In terms of personal development Ive beginning to make some strides doing SQL queries and also acquired basic working knowledge of the Knime software for data extraction and transformation. So far it has been a more stable platform for database manipulation in order to optimize Tableau dashboards, which suffer in performance when dealing with massive amounts of data. Goal for the end of the year is to automate Knime workflows as currently I need to process them manually to refresh the data. I've also takin all the mandatory training in time.

De Leon Melendez, Jesus Manuel 1/23/2023

Best Network-Customer Focus

Best 4G Network: Reclaim Root 4G overall performance and reliability wins. Offload 4G traffic to c-band to improve 4G capacity and performance.

Capacity and performance management

SGFI, CDMAless Get Well Plan & Top 10 trouble areas

ROOT Metrics Metro Wins	75%
ROOT Metrics State Drives	TBD
% 4G Connections on 4G	99%
Paging Success	97%
VoLTE Paging Success Rate	98%
WLS % Triggering Sectors	5.9%*
WLS % VoLTE Ineffective Attempts	0.6%
WLS % VOLTE Lost Calls	0.6%
VoLTE RTP Gap Duration	0.11%

LTE RRC Setup Failure	0.25%
LTE Bearer Drop	0.40%
Total 4G/5G Build	2,666
% of eNB Parameters Under Management	90%
% of Sites Deployed (V-SON & 3rd Party)	85%
SNAP 4G/5G Sub 3 GHz Sites Converted	11,096
LTE Home Drop Rate	1%
LTE Home Setup Failure Rate	1%
LTE Home Latency (80/80)	50 ms
LTE Home DL Throughput (80/80)	25 Mbps
LTE Home UL Throughput (80/80)	4 Mbps
*Top 46 5.2 to 3.3%	
30 4.7 to 1.9-2.6%	
All Other 9.1 to 9.8%	

Best 5G Network: Measurable and demonstrated

Maintain Win in Root Most Reliable 5G Network. 5G C-band launch and continued deployment. Stretch goal of 200M POPs by end 2022. 5G mmWave continued deployment, 5G low band continued deployment via One Touch

In-building & Private 5G

5G C-band Sites On Air LTD	23,000
5G C-band Sites Built Capable (Early 30 PEAs)	4,368
5G UW POPs Covered	200M
5G C-band Throughput	100 Mbps
5G on 5G (Dual Connect Rate)	Tracking
5G UW Setup Failure Rate	1.0%
5G UW Drop Rate	5.0%
5G UW Throughput	200 Mbps

5G mmWave Sites On Air	12,006
SNAP 5G mmWave Sites Converted	2,852
5G mmWave Throughput	300 Mbps
5G mmWave Setup Failure Rate	1%
5G Nationwide Setup Failure Rate	1%
5G Nationwide Drop Rate	10%
5G FWA % Data Volume on 5G	90%
5G FWA Drop Rate	1%
5G FWA Setup Failure Rate	1%
5G FWA Latency (80/80)	20 ms
5G FWA DL Throughput (80/80)	300 Mbps
5G FWA UL Throughput (80/80)	50 Mbps
5G FWA mmWave Households Covered	9.897M
Enable Public MEC Cities (incremental)	10

In terms of maintaining our leadership as the best network, everything we do in the team is focus on that. I've held capacity meetings with the regions to evaluate solutions and educate them on why and how we are prioritizing solutions. Ive also lead meetings to discuss both Root Metric initiatives and answer any questions they might have.

I've Worked with Abdul developing the 5G UWB combo dashboard and on the logic for the sall cell opportunities based on the RF Standards team guidance. Updated the Solutions insider Dot Metrics dashboard and added Automatica classification based on SPM project coordinates.

De Leon Melendez, Jesus Manuel 7/26/2022

see attached file

Watson, James E 1/26/2023

Overall Assessment

Overall Performance Summary

Throughout the 2nd half of 2022 Jesus worked with Gene on new dashboard to focusing on which triggers to solve. Jesus was able to use tableau to test and show the various permutations of testing for Gene's view of the prioritization of which trigger(s) to solve.

On 10/5 – accepted ownership of RFDS mismatch dashboard to provide updates. This dashboard is an important dashboard as it highlights data integrity issues which the regional teams need to work on. Jesus updating the dashboard for the Data Integrity team supports the team's effort to clean up issues in the databases.

Continued to support KRW calculation in NPP by doing testing and evaluation of the results. Throughout the 2nd half of 2022 there were a number of issues with the 'K' or Kracken value in KRW. The calculation to derive the 'K' were tested by Jesus and Mike on a couple of occasions to ensure accuracy. Later there were many issues and subsequent testing required for KSP and KSP (which uses the 'K' value). Jesus worked with Mike and provided the testing needed in a timely manner.

Worked to develop a program labels dashboard which would use logic to determine which solutions needed Sub1, Sub3 or both labels. That simplified the team's effort to correctly attach program labels to the thousands of existing projects already released. As we worked through the task of updating the program labels to those solutions Jesus made some updates/improvements to the logic to improve the process. The dashboard was valuable in that a) it provided a consistent application of those labels and b) it saved the entire team time rather than everyone try to figure it out on their own.

Watson, James E 1/30/2023

Year End Performance Rating

Performing

Attachments

File Name	Uploaded By	Uploaded Date
EOY2022 KPIs (1).xlsx	Watson, James E	01/19/2023

Manager Signature:	Watson, James	Date:	1/30/2023
---------------------------	---------------	--------------	-----------

Signing does not indicate you agree with the document, only that the document was reviewed with you.

Employee Signature: De Leon Melendez, Jesus
Manuel_____

Date: 1/30/2023